

Clinical Natural Language Technology for Healthcare: A Hybrid Retrieval-Augmented Clinical Document Intelligence System for Cotiviti

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Abstract

Most clinical knowledge still lives in unstructured documents—discharge summaries, operative notes, payer correspondence—where it is difficult to search and expensive to review manually. This report surveys how clinical Natural Language Processing has evolved from rule-based systems to Large Language Models (LLMs), identifies the opportunities and risks this shift creates for Cotiviti, and presents **ClinicalDoc AI**: a working proof-of-concept that uses hybrid Retrieval-Augmented Generation (RAG) to answer clinical questions with page-level source citations. The system combines BM25 sparse retrieval, FAISS dense vector search, and Reciprocal Rank Fusion, grounding every LLM response in the actual document rather than parametric memory. As part of this assessment, a working proof-of-concept named ClinicalDoc AI was developed to demonstrate how Hybrid Retrieval-Augmented Generation can support intelligent clinical document analysis. Rather than relying solely on LLM reasoning, the system grounds every response using retrieved medical evidence with page-level citations, making it suitable for healthcare document review workflows.

1. Introduction

Cotiviti’s core business—payment integrity, utilization management, and clinical documentation review—depends on extracting accurate information from medical records at scale. Today, that work is largely manual: reviewers read multi-page PDFs, locate relevant passages, and render clinical judgements one chart at a time. The bottleneck is not clinical expertise; it is information retrieval (Topol, 2019). Clinical NLP has the potential to eliminate this bottleneck, but only if the technology can be trusted to retrieve the right evidence and present it transparently. This report traces the technology’s evolution, highlights where Cotiviti stands to gain the most, and demonstrates a concrete system—ClinicalDoc AI—built specifically to address these needs.

2. Current Trends in Clinical NLP

Clinical NLP has moved through three distinct phases. Early systems relied on handwritten rules and medical ontologies like SNOMED-CT and ICD-10. These were precise but brittle—any new phrasing or abbreviation required manual rule updates. The second phase introduced statistical models: BM25 for lexical search and later transformer architectures like GatorTron, which could learn contextual meaning from large clinical corpora (Yang et al., 2022).

The current phase is defined by general-purpose LLMs. Models such as GPT-4 and Med-PaLM 2 have achieved near-physician performance on medical licensing exams (Singhal et al., 2023), demonstrating strong potential for summarization, question answering, and coding assistance. However, LLMs generate answers from parametric memory, which means they can produce confident but factually wrong outputs—a phenomenon called hallucination (Thirunavukarasu et al., 2023). Retrieval-Augmented Generation addresses this directly by forcing the model to ground its answers in retrieved evidence rather than relying on training data alone (Lewis et al., 2020). Recent work shows that hybrid RAG—combining sparse lexical and dense semantic retrieval—consistently outperforms either method in isolation (Gao et al., 2024).

3. Industry Challenges

Despite these advances, deploying clinical NLP in production remains difficult for several practical reasons. Hallucination is the most discussed risk, but in a payment-integrity context the downstream cost is concrete: a hallucinated diagnosis code can trigger an incorrect claim decision. Regulatory requirements compound the problem—HIPAA mandates strict controls over protected health information (PHI), and any AI system that processes clinical text must produce auditable outputs with clear provenance. Beyond compliance, clinical documents are inherently heterogeneous: different hospitals use different templates, abbreviation conventions, and scanning quality, which means a model trained on one health system’s data may not generalize well to another. Finally, most claims adjudication platforms were built before modern NLP existed, so integration requires careful API design rather than simple plug-and-play deployment.

4. Opportunities for Cotiviti

These challenges are precisely where Cotiviti is well-positioned to act. In **clinical documentation review**, a hybrid RAG system can surface the exact passages a reviewer needs—diagnosis, medications, lab values—without requiring them to read every page of a chart. In **payment integrity**, the same retrieval pipeline can cross-reference billing codes against clinical evidence in the record, flagging discrepancies for auditor review. For **medical coding assistance**, source-cited question answering helps coders verify that documentation supports the assigned codes. And in **utilization management**, automated summarization can compress lengthy clinical records into structured overviews, accelerating authorization decisions. The key insight is that a single hybrid retrieval architecture can serve multiple Cotiviti workflows—the difference is the query and the document, not the underlying system.

5. Proposed Solution: ClinicalDoc AI

ClinicalDoc AI is a working prototype that implements hybrid RAG for clinical document intelligence. The architecture (Figure 1) has four stages:

1. **Ingestion.** Medical PDFs are parsed with PyMuPDF, cleaned (whitespace normalization, non-printable character removal), and split into 1,000-character chunks with 200-character overlap using LangChain’s RecursiveCharacterTextSplitter. Each chunk retains its source filename and page number.
2. **Dual Indexing.** Every chunk is indexed twice: once in a FAISS vector store using OpenAI text-embedding-3-small embeddings (dense retrieval), and once in a BM25Okapi index (sparse lexical retrieval) (Johnson et al., 2021).
3. **Hybrid Retrieval.** When a user asks a question, the system first generates four semantically diverse reformulations via an LLM-powered multi-query generator. Each variant is run against both indexes, and the combined results are merged using Reciprocal Rank Fusion ($k=60$), which scores every document based on its rank across all result lists (Cormack et al., 2009).
4. **Grounded Generation.** The top- k fused passages are fed as context to GPT-4.1-mini with a prompt that explicitly forbids outside knowledge and requires page-level citations in every answer.

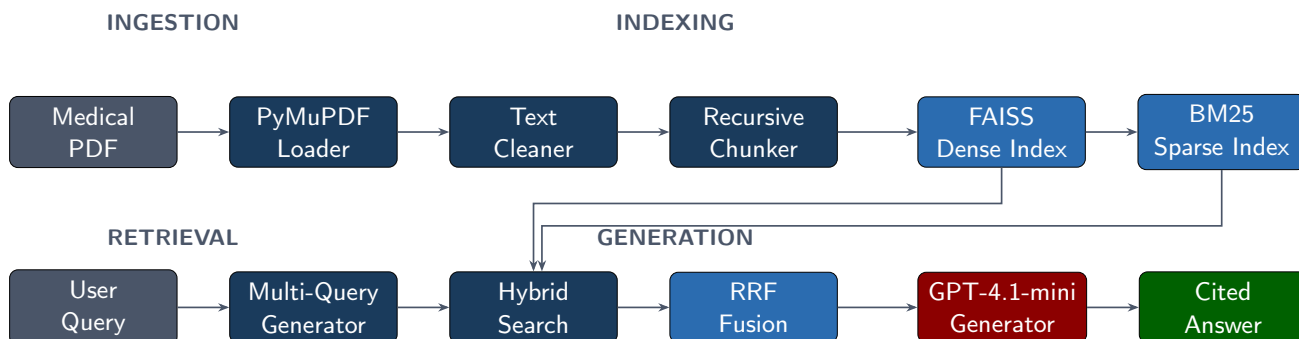


Figure 1: ClinicalDoc AI architecture — four-stage hybrid RAG pipeline from document ingestion to cited answer generation.

The front end is a Streamlit application that lets users upload a PDF, ask free-text clinical questions, generate full-

document summaries, and inspect the source passages behind every answer. The interface also tracks conversation history and allows export, supporting the kind of audit trail that Cotiviti’s compliance workflows require.

Table 1: Traditional clinical documentation review vs. ClinicalDoc AI.

Dimension	Traditional Workflow	ClinicalDoc AI
Information access	Manual page-by-page reading	Dual-indexed hybrid search
Query resolution	Reviewer memory and keyword search	Multi-query retrieval with RRF fusion
Answer provenance	Reviewer notes (limited audit)	Automatic page-level source citations
Summarization	Manual chart abstraction	LLM-generated structured summary
Throughput	Constrained by reading speed	Sub-second retrieval per query
Consistency	Varies with reviewer experience	Deterministic retrieval; reproducible
Transparency	Evidence verification depends on reviewer	Automatic page-level evidence with source citations

6. Business Value

The practical value of ClinicalDoc AI comes from three properties. First, *hybrid retrieval* catches both exact clinical terms (via BM25) and semantically related passages (via FAISS), which means reviewers are less likely to miss relevant evidence. Second, *source citations* attached to every answer create a built-in audit trail—something Cotiviti already requires in its documentation review and payment integrity workflows. Third, the architecture is *modular*: the retriever, index, and generator are separate components, so the system can be adapted to different document types—clinical notes, billing policies, payer contracts—without rebuilding the pipeline.

7. Risks and Limitations

- **Retrieval failure.** If the chunking boundaries split a critical sentence, the retriever may return incomplete context, leading to partial or incorrect answers. Tuning chunk size and overlap mitigates this but does not eliminate it.
- **Residual hallucination.** RAG reduces hallucination substantially, but the LLM can still misinterpret retrieved passages. In a clinical setting, every generated answer should be reviewed by a qualified professional before acting on it.
- **PHI exposure.** The current prototype sends document text to an external API (OpenAI). A production deployment would need either a self-hosted model or a BAA-covered API endpoint to satisfy HIPAA requirements.
- **Model maintenance.** LLM providers update and deprecate models on their own schedule. Cotiviti would need version-pinning and regression testing to prevent silent behaviour changes.

8. Future Scope

The most impactful next steps would be: (a) fine-tuning a domain-specific embedding model on Cotiviti’s clinical corpora to improve retrieval precision for specialized terminology, (b) adding ICD-10 and CPT entity extraction to support automated coding validation, (c) implementing PHI de-identification and role-based access control for HIPAA-compliant deployment, and (d) extending the pipeline to multi-document comparison, enabling side-by-side analysis of clinical notes against billing policies or practice guidelines.

9. Conclusion

Clinical NLP has reached a point where LLMs can meaningfully assist—not replace—human reviewers in document-intensive healthcare workflows. ClinicalDoc AI demonstrates one practical way to do this: by combining BM25, FAISS, multi-query expansion, and Reciprocal Rank Fusion into a hybrid retrieval layer that grounds every LLM response in the source document. For Cotiviti, this approach directly supports payment integrity, clinical documentation review, and coding assistance while maintaining the traceability and auditability that regulated healthcare demands.

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